

PhD Qualification Exam in Applied Mathematics

Mock Examination No. 5

Total: 100 points

Problems 1–5 are required. Solve exactly one of Problems 6 and 7. If both are attempted, only the higher score will be counted.

- 1. (15 points)** Construct a two-node quadrature rule on $[0, 1]$ that is required to include the left end-point and has maximal degree of exactness:

$$Q(f) = w_0 f(0) + w_1 f(c), \quad 0 < c < 1.$$

Determine c, w_0, w_1 , prove maximality of the degree, and derive the error formula for $f \in C^3[0, 1]$, including the error sign when $f^{(3)} \geq 0$.

- 2. (15 points)** Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ have a zero x_* , with $J(x_*)$ nonsingular and J locally Lipschitz. An inexact Newton step satisfies

$$J(x_k)s_k = -F(x_k) + r_k, \quad x_{k+1} = x_k + s_k, \quad \|r_k\|_2 \leq \eta_k \|F(x_k)\|_2.$$

Derive a local error inequality. Prove local linear convergence if $\sup_k \eta_k < 1$ is sufficiently small, super-linear convergence if $\eta_k \rightarrow 0$, and quadratic convergence if $\eta_k = O(\|F(x_k)\|_2)$.

- 3. (20 points)** Let $A \in \mathbb{R}^{m \times n}$, $m \geq n$, have full column rank, and define

$$K = \begin{bmatrix} I_m & A \\ A^T & 0 \end{bmatrix}.$$

- (a) Show that solving $K \begin{bmatrix} r \\ x \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix}$ is equivalent to the least-squares problem $\min_x \|Ax - b\|_2$.
- (b) In terms of the singular values $\sigma_i(A)$, determine all eigenvalues of K , including multiplicities. Deduce that K is symmetric, nonsingular, and indefinite.
- (c) Choose an appropriate Krylov method among CG, MINRES, and GMRES, justify the choice, and state the matrix-vector cost and a natural block preconditioning strategy.

- 4. (15 points)** For the heat equation $u_t = u_{xx}$, consider Crank–Nicolson on a periodic grid:

$$\frac{U_j^{n+1} - U_j^n}{\tau} = \frac{1}{2} \delta_{xx}(U_j^{n+1} + U_j^n), \quad r = \frac{\tau}{h^2}.$$

Determine its consistency order and Fourier amplification factor, and prove unconditional ℓ^2 stability. Show that high-frequency amplification approaches -1 as $r \rightarrow \infty$ and explain why this may produce visible temporal oscillations despite stability. State one standard remedy.

- 5. (15 points)** For $-\varepsilon u'' + au' = f$ on $(0, 1)$ with $a > 0$, compare centered and upwind discretizations on a uniform grid.

- (a) Derive the centered matrix and prove that its discrete maximum principle requires the cell-Peclet condition $ah \leq 2\varepsilon$.
- (b) Derive the backward-upwind matrix and prove monotonicity for every $h > 0$.
- (c) Determine the leading truncation error of the upwind derivative and interpret it as numerical diffusion.

6. (20 points; choose one of Problems 6–7) For fixed detuning $\sigma \in \mathbb{R}$, consider

$$y'' + y = \varepsilon \cos((1 + \varepsilon\sigma)t), \quad y(0) = y'(0) = 0.$$

Use multiple scales to construct an approximation valid for $t = O(\varepsilon^{-1})$. Describe the limit $\sigma \rightarrow 0$, compare with the exact solution, and state the accuracy on the slow time interval $0 \leq \varepsilon t \leq T$.

7. (20 points; choose one of Problems 6–7) Consider the primal-dual pair

$$\min c^T x \quad \text{s.t. } Ax = b, x \geq 0, \quad \max b^T y \quad \text{s.t. } A^T y + s = c, s \geq 0,$$

where A has full row rank and strictly feasible primal and dual points exist.

- (a) Derive the central-path equations and prove that the duality gap is $n\mu$.
- (b) Linearize the central-path residual equations and eliminate variables to obtain a symmetric positive-definite Schur-complement system.
- (c) State one admissible stopping criterion and the dense arithmetic cost of a Newton step.